

Guide 1. APT Project Definition Capstone Course

A. PART I

1. Personal Background

Below is a table where you must fill in the requested information.

Student name	Cristian Olivares Constanza Contreras
RUT	19.770.359-8 21.138.315-1
Program	Computer Engineering
Campus	Viña del Mar

2. APT Project Description

In the description, briefly state the name of your APT project and the graduate profile competencies you will apply. If your program has defined performance areas, also mention which performance areas the project is linked to.

Project name	<i>MetalConnect</i>
Performance area(s)	<i>Systems analysis and design, Cybersecurity, Software development, Project management.</i>
Competencies	<i>Develop software solutions using agile methodologies; Manage relational and non-relational databases; Implement distributed systems architectures; Apply good information security practices.</i>

3. APT Project Rationale

Below are different fields that you must complete with the requested information. This section seeks for you to describe your project in detail and justify its relevance and pertinence.

Relevance of the APT Project	<i>Chile's metalworking and industrial sector lacks specialized digital platforms that efficiently connect workshops, independent tradespeople, and clients who need machining, welding, fabrication, and other services in the field. This project was conceived with nationwide scope, and seeks to close the existing technological gap in this productive sector. It impacts small and medium-sized metalworking businesses, independent workers in the field, and clients who require specialized industrial services. MetalConnect's value contribution is to digitize and facilitate the connection between supply and demand for metalworking services, reducing search times and improving the visibility of service providers.</i>
APT Project Description	<i>MetalConnect is a mobile application built with React Native and Expo that functions as a digital marketplace for Chile's metalworking and industrial sector. Its goal is to connect metalworking workshops, independent tradespeople, and clients who need industrial services. The application allows clients to post jobs with a description and photos, while workshops and independent workers can view these requests and contact the client directly via WhatsApp. It includes a profile system, a portfolio of completed jobs, anonymous ratings, and an artificial intelligence feature based on the Gemini API to assist users. The backend is implemented on Supabase, with authentication, a PostgreSQL database, and file storage.</i>
Relevance of the Project to the Graduate Profile	<i>MetalConnect is directly related to the graduate profile of DuocUC's Computer Engineering program, since it requires applying competencies in mobile software development, database administration, API implementation, and secure data management. The project integrates technologies such as React Native, Supabase, and AI services, making it possible to put into practice the competencies in systems architecture design, application development, and the application of good information security practices acquired throughout the program.</i>
Relationship to Professional Interests	<i>Our professional interest is centered on developing mobile applications and technology solutions with a real impact on society. MetalConnect directly reflects these interests, as it allows me to develop a complete mobile application from scratch, applying modern technologies such as React Native, Expo, and Supabase. In addition, the project represents a real challenge for an actual client, which will allow me to gain experience developing digital products for the job market. Carrying out this project contributes to my professional development by tackling real problems in software architecture, data security, API integration, and publishing to app stores.</i>
Feasibility of Developing the APT Project	<i>The project is feasible to develop considering: (1) The semester is long enough to complete the MVP planned within 6 weeks. (2) The hours allotted allow progress to be made week by week on each feature. (3) The required resources are accessible: a personal computer, a free Supabase account, a free Expo account, and a mobile device for testing. (4) Factors that facilitate development: having a real client (Hidrameec)</i>

	<i>and access to official documentation for all the technologies. (5) Factors that could pose difficulties: the learning curve for new technologies such as React Native and Supabase, and possible compatibility conflicts between libraries. Solution: dedicate time to research and follow the official documentation.</i>
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B. PART II

4. Objectives

In this section you must define the general and specific objectives of the APT Project. It is important to clarify that the objectives should be stated clearly, concisely, and without further explanation — that is, they should be self-explanatory. It is suggested that they be written using a verb in the infinitive, since this forces you to specify concrete actions.

General objective	<i>Develop a mobile application (Android) that connects metalworking workshops, independent tradespeople, and clients who require industrial services in Chile, through a digital marketplace with features for posting jobs, user profiles, a ratings system, and direct contact via WhatsApp.</i>
Specific objectives	<i>1. Implement a secure authentication system with registration, login, and password recovery using Supabase Auth. 2. Develop the job-posting module that allows clients to describe their requirements and attach photos. 3. Create differentiated profiles for clients, workshops, and independent workers, with a portfolio and an anonymous ratings system. 4. Integrate an artificial intelligence feature using the Gemini API to assist users in describing jobs. 5. Publish the application on the Google Play Store at the end of the development period.</i>

5. Methodology

In the following section you must describe the methodology, specific to your discipline, that you will use to carry out the APT project described above, including the stages and working methods.

Description of the Methodology

An agile development methodology is used, working in weekly sprints and dividing the project into 6 weeks: Week 1: Home screen and project setup. Week 2: Supabase implementation, authentication, and user registration. Week 3: Development of the client area with job posting. Week 4: Development of the workshop and independent-worker area. Week 5: AI integration with the Gemini API and the ratings system. Week 6: Bug fixing and publication on the Google Play Store. Development team: Cristian Olivares - Backend and database (Supabase, authentication, business logic, security). Constanza Contreras - Frontend and interface design (React Native, Expo, views, and visual components).

6. Evidence

Below, describe which evidence will be assessed in the progress report and in the final report of your APT project. This evidence must be agreed upon with your instructor. Evidence is understood as the products developed during the project whose purpose is to make visible or document how the work has been implemented.

Type of evidence (progress or final)	Evidence name	Description	Justification
	METALCONNECT README	Evidence of the MetalConnect app's README, including the start date and the components it contains	Demonstrates the real progress of the project's development in a documented and verifiable way

7. Work Plan

In the following table, define the planning of your APT Project according to what is required.

APT Project Work Plan						
Competency or competency units	Name of Activities/Tasks	Description of Activities/Tasks	Resources	Activity duration	Responsible ¹	Comments
Project management	Kick-off meeting with the client	Project presentation and alignment of expectations	Video call/WhatsApp	1 week	Cristian Olivares/Constanza Contreras	-
Database design and modeling	Design and configuration of the database in Supabase	Creation of the database schema (profiles, jobs, portfolio_items), authentication functions, and triggers	Supabase, documentation	2 weeks	Cristian Olivares	Adjustment of RLS policies for role permissions
Implement distributed systems	Architecture design	Define the communication between the mobile application, backend, database, and external services.	Supabase, React Native, Expo	1 week	Constanza Contreras	Good communication between components must be ensured
Design and systems analysis	Creation of mockups	Design of main screens	Figma	1 week	Cristian Olivares/c	-
Develop software solutions	Application development	Program the application's main functions	Visual Studio/Cursor, Expo, React Native.	11 weeks	Cristian Olivares/Constanza Contreras	Errors may appear during programming and integration.
Implement AI	AI and Chatbot integration	Incorporate AI tools to support quotes and a chatbot.	Gemini AI and chatbot API.	1 week	Constanza Contreras	-
Apply security in the system	Security implementation	Incorporate authentication, access controls,	MobSF	1 Week	Constanza Contreras	The necessary tests must be carried out to

¹ If the APT Project is a group project, use this column to indicate the name of the person responsible for each task or activity. This will later make it possible to differentiate the assessment for each team member.

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